If we are to send people, it must be for a very good reason—and with a realistic understanding that almost certainly we will lose lives. Astronauts and cosmonauts have always understood this. Nevertheless, there has been and will be no shortage of volunteers.

But why Mars? Why not return to the Moon? It's nearby, and we've proved we know how to send people there. I'm concerned that the Moon, close as it is, is a long detour, if not a dead end. We've been there. We've even brought some of it back. People have seen the Moon rocks, and, for reasons that I believe are fundamentally sound, they are bored by the Moon. It's a static, airless, waterless, black-sky, dead world. Its most interesting aspect perhaps is its cratered surface, a record of ancient catastrophic impacts, on the Earth as well as on the Moon.

Mars, by contrast, has weather, dust storms, its own moons, volcanos, polar ice caps, peculiar landforms, ancient river valleys, and evidence of massive climatic change on a once-Earthlike world. It holds some prospect of past or maybe even present life, and is the most congenial planet for future life—humans transplanted from Earth, living off the land. None of this is true for the Moon. Mars also has its own legible cratering history. If Mars, rather than the Moon, had been within easy reach, we would not have backed off from manned space flight.

Nor is the Moon an especially desirable test bed or way station for Mars. The Martian and lunar environments are very different, and the Moon is as distant from Mars as is the Earth. The machinery for Martian exploration can at least equally well be tested in Earth orbit, or on near-Earth asteroids, or on the Earth itself—in Antarctica, for instance.

Japan has tended to be skeptical of the commitment of the United States and other nations to plan and execute major cooperative projects in space. This is at least one reason that Japan, more than any other spacefaring nation, has tended to go it alone. The Lunar and Planetary Society of Japan is an organization representing space enthusiasts in the government, universities, and major industries. As I write, the Society is proposing to construct and stock a lunar base entirely with robot labor. It is said to take about 30 years and to